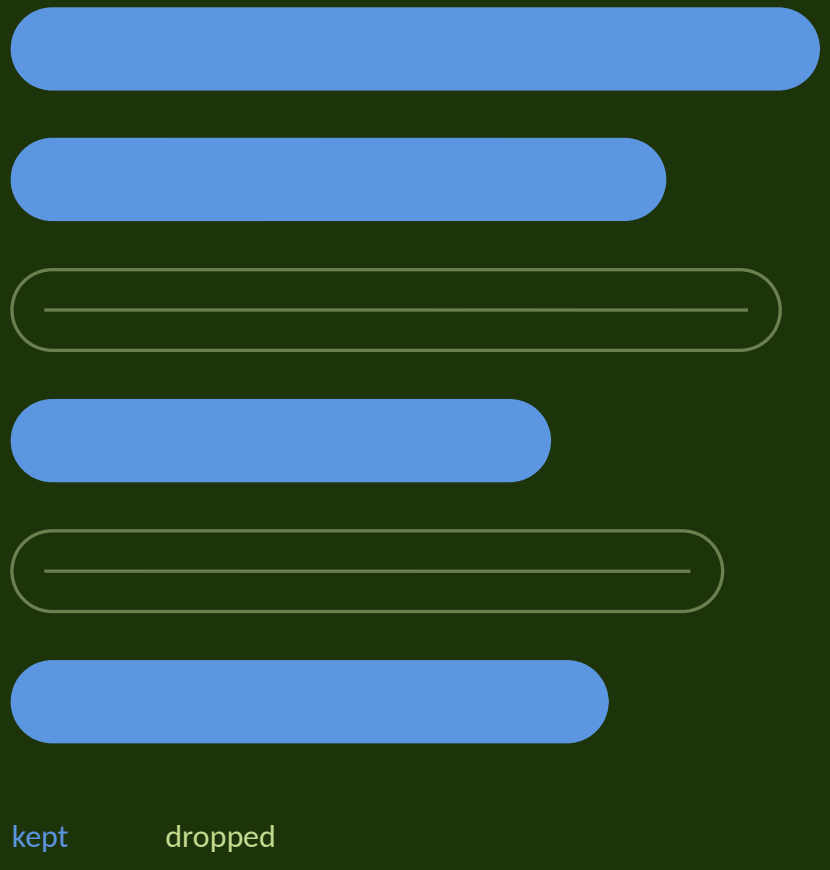
Conformal Guard

Lets a pharmacist bound how often Muse Spark's unchecked medical answers contradict the reference answer, and says how many drafts they must read



Meta Global AI Developer Hackathon 2026 | Indie developer track | Theme: Build for Your World

Confident answers, unknown error rate

Wrong facts sound like right ones

A model states a wrong dose or interaction in the same calm tone as a correct one.

No token probabilities

Muse Spark's API rejects logprobs, so any signal has to come from the text alone.

A score is not a promise

Detectors output a number. Nobody can say which threshold is safe, or how often it fails.

6%

of Muse Spark answers contain a sentence that contradicts the MedLFQA reference answer

Training and calibration answers. Held-out test set: 4.5%.

What the guarantee says

At least 95% of filtered answers contain no sentence that contradicts the MedLFQA reference answer

α = 0.05

Averaged, not per answer

It holds on average over questions like these, not for one answer or for the ones the system sends to review.

Relative to the labels

Judged by an NLI model against MedLFQA's reference answer, which a doctor wrote for 14% of test questions.

New questions like the old

It assumes exchangeability. A similarity check flags clear outliers, mostly very short questions.

How it works

Generate

Muse Spark 1.3 answers once, then 5 more times with the same settings

Split

A fixed rule-based splitter turns the answer into sentences

Score

Agreement with resamples (NLI), MedlinePlus evidence (FAISS + MiniCheck), position, length

Calibrate

Threshold τ set on 328 answers the score never saw

Decide

Ship, send for review or block, as a LlamaFirewall scanner

Train · 246 answers
fit the score combiner

Calibrate · 328 answers
set τ and the ship threshold

Test · 245 answers
report every number

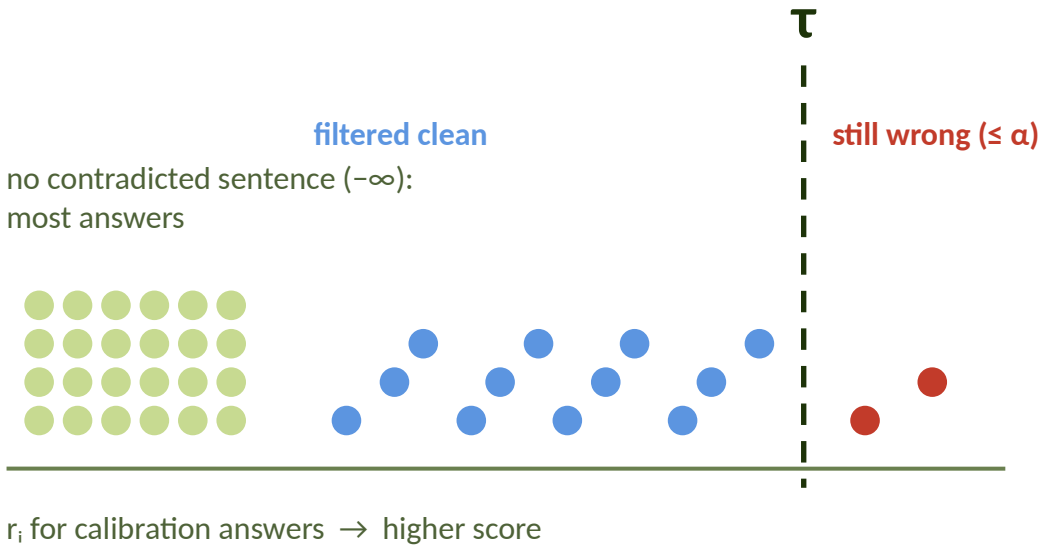
Why the threshold works

1. For each calibration answer i, let r_i be the highest score among its contradicted sentences ($-\infty$ if it has none).

2. Set τ to the k-th smallest r_i, with $k = \lceil (n+1)(1-\alpha) \rceil$.

3. Keep only sentences scoring above τ. A new answer comes out clean exactly when its $r \le \tau$.

4. By exchangeability, $P(r \le \tau) \ge k/(n+1) \ge 1-\alpha$, whatever the score.



Ship rule: an answer goes out unchanged if its weakest sentence clears λ, chosen by conformal risk control so that P(shipped and contradicted) ≤ α.

Data and labels, checked by hand

819

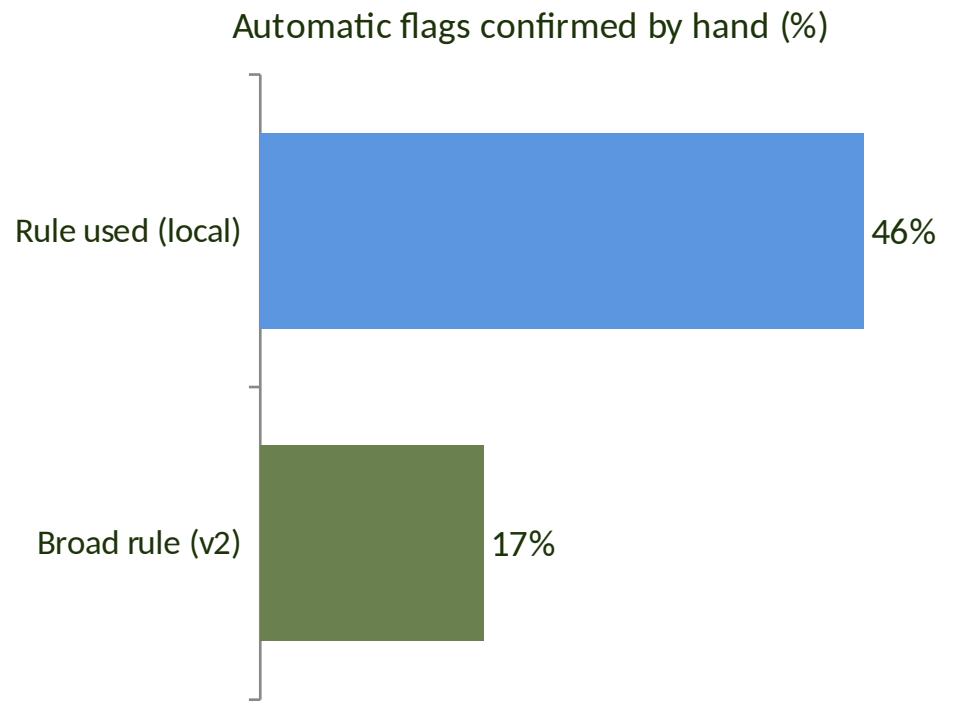
MedLFQA questions in three groups: clinical, consumer, medication

4,914

Muse Spark generations (Oct 26, 2026): one answer and five resamples each

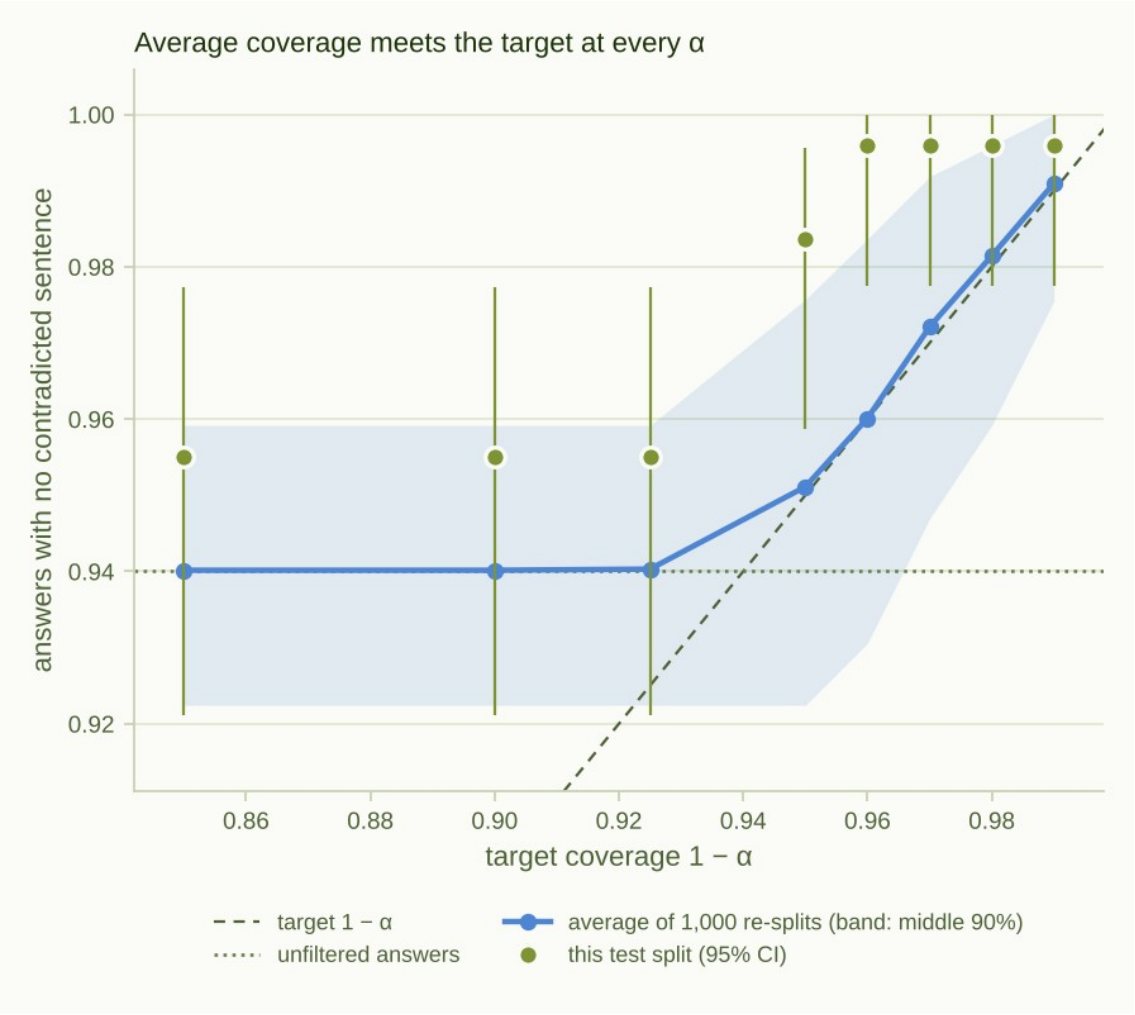
4,840

sentences, each labelled by NLI against the MedLFQA reference answer



10 of 59 broad flags (17%) and 6 of 13 flags under the rule used (46%, 95% CI 19 to 75%) were real disagreements; 0 of 15 unflagged sentences were. The rule was chosen on the training split, before any calibration result.

The guarantee held on held-out answers



98.4%

of 245 test answers clean (95% CI 95.9 to 99.6%), against a 95% target; 95.5% before filtering

95.1%

average over 1,000 re-splits, as theory predicts: a check that the code is right

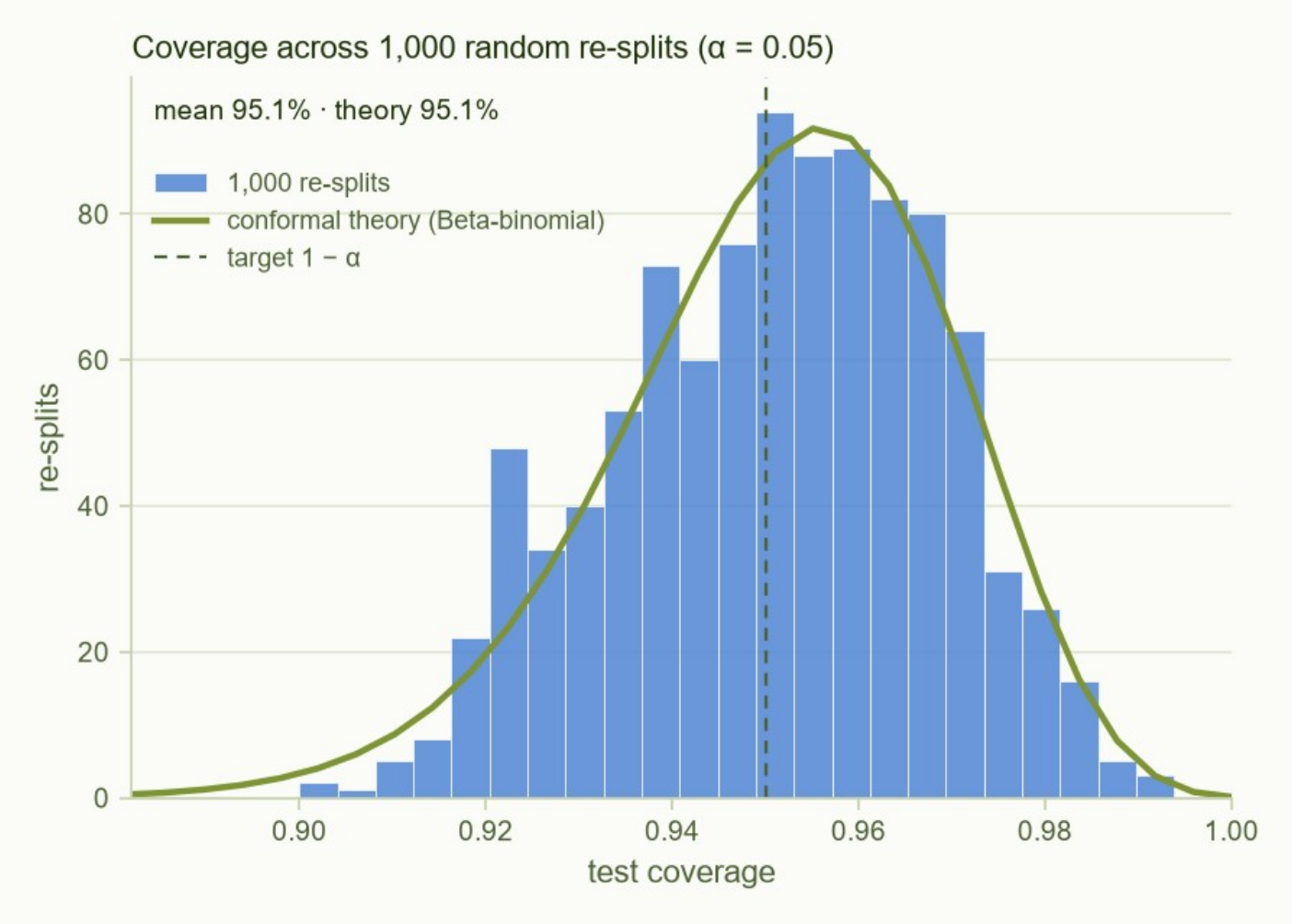
Unfiltered answers are already 94–95.5% clean, so at α = 0.05 the filter has little to do. Where they meet the target, it does nothing.

A check that the code is right

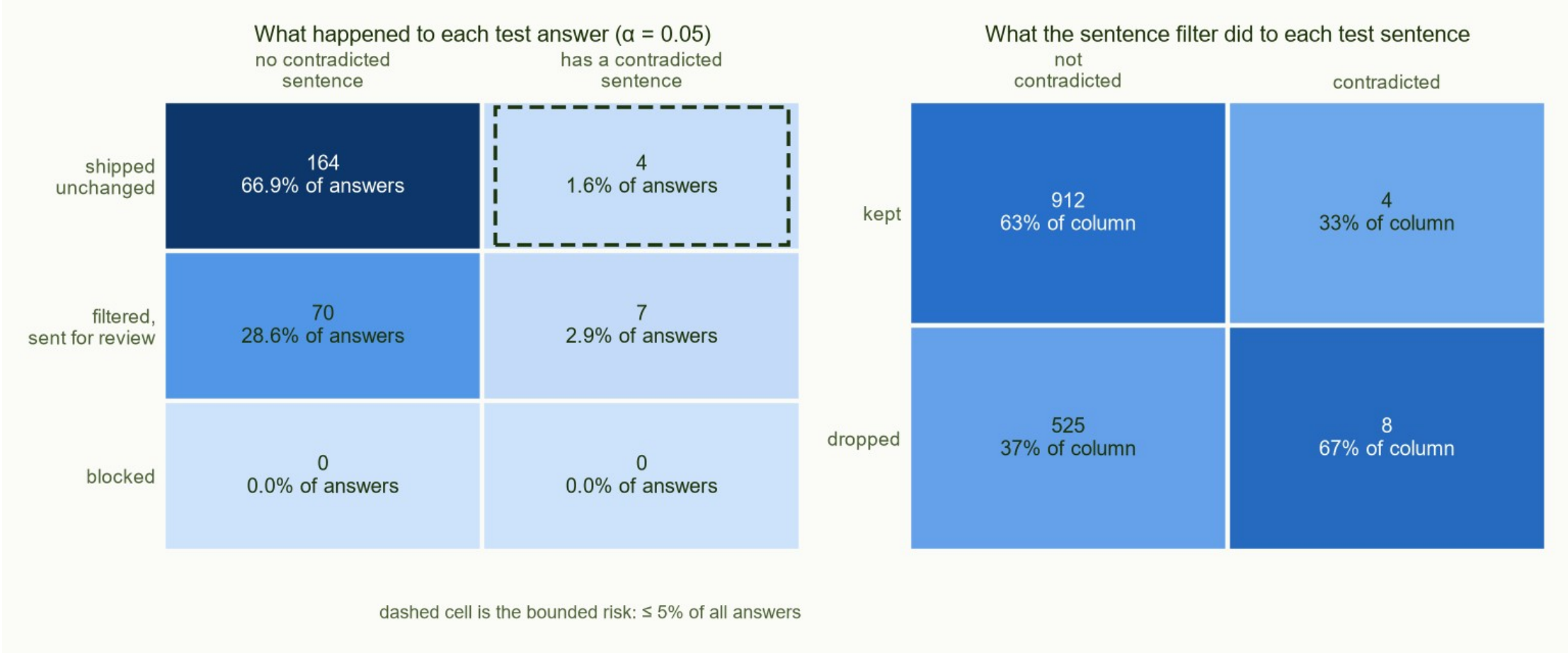
Any single test set can land anywhere between 92% and 98%. That spread is predicted too.

Coverage given a calibration set follows a Beta distribution, so across re-splits it is Beta-binomial. The green curve is that prediction, not a fit.

Any correct conformal filter matches it, even with a random score: this checks the code, not the method.

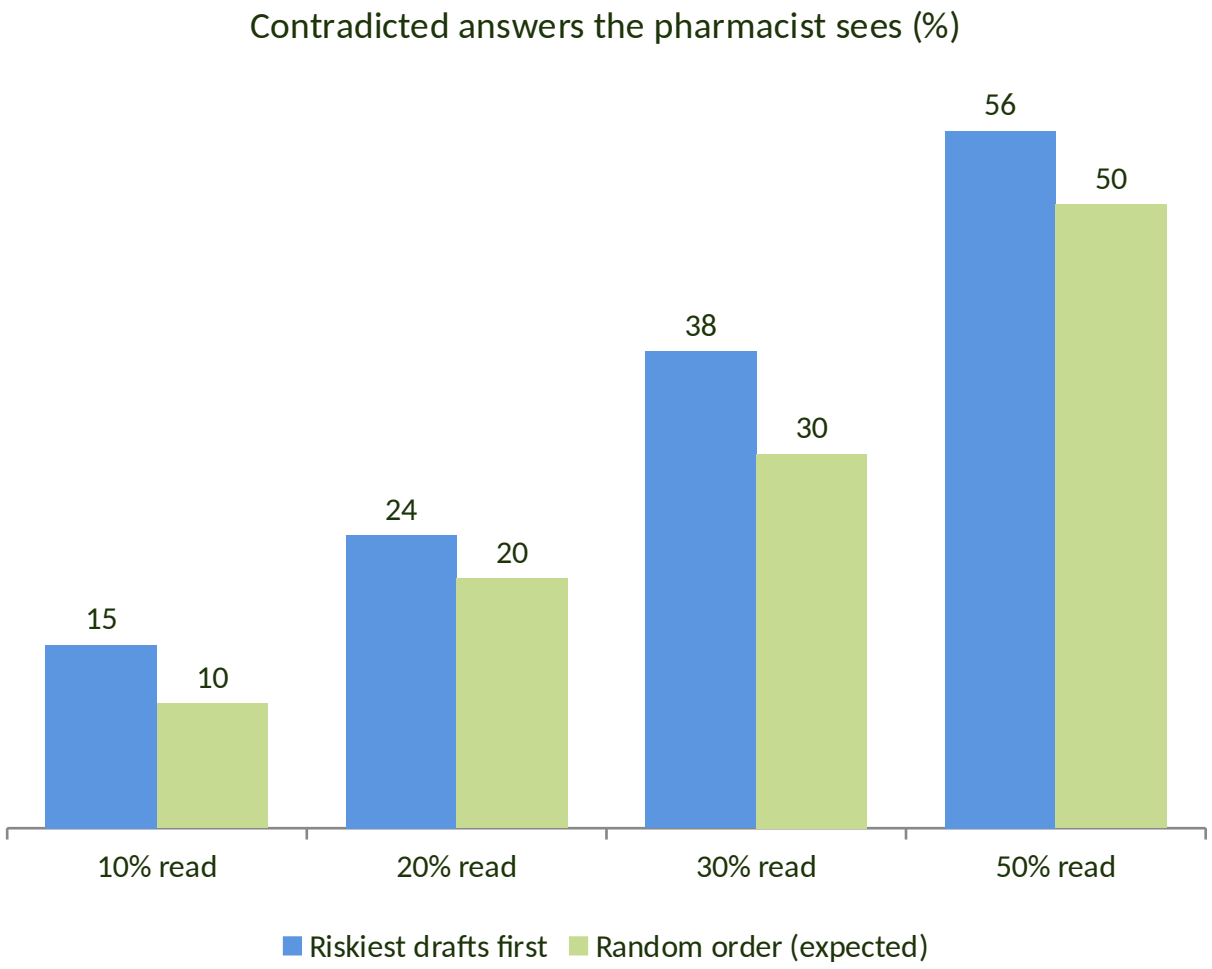


What happened to every test answer



4 of 245 answers (1.6%) were shipped with a contradicted sentence, against a 5% bound. On this split, reviewed answers looked riskier (9.1% vs 2.4% of shipped); the next slide shows that didn't hold up. The filter dropped 8 of 12 contradicted sentences and 37% of fine ones.

The bound held up. The triage didn’t



4.7% ≤ 5%

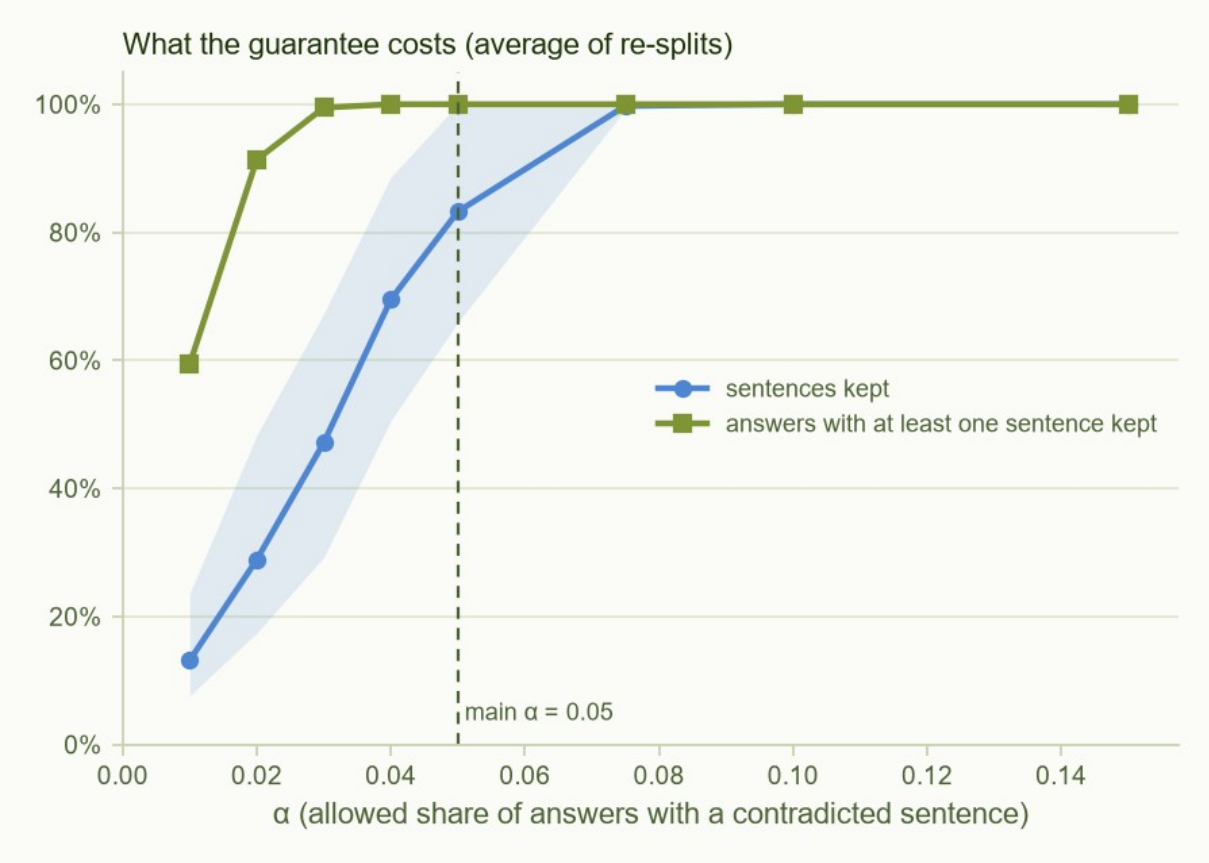
of answers shipped with a contradiction, averaged over 1,000 re-splits of all 573 held-out answers (5.9% without the guard)

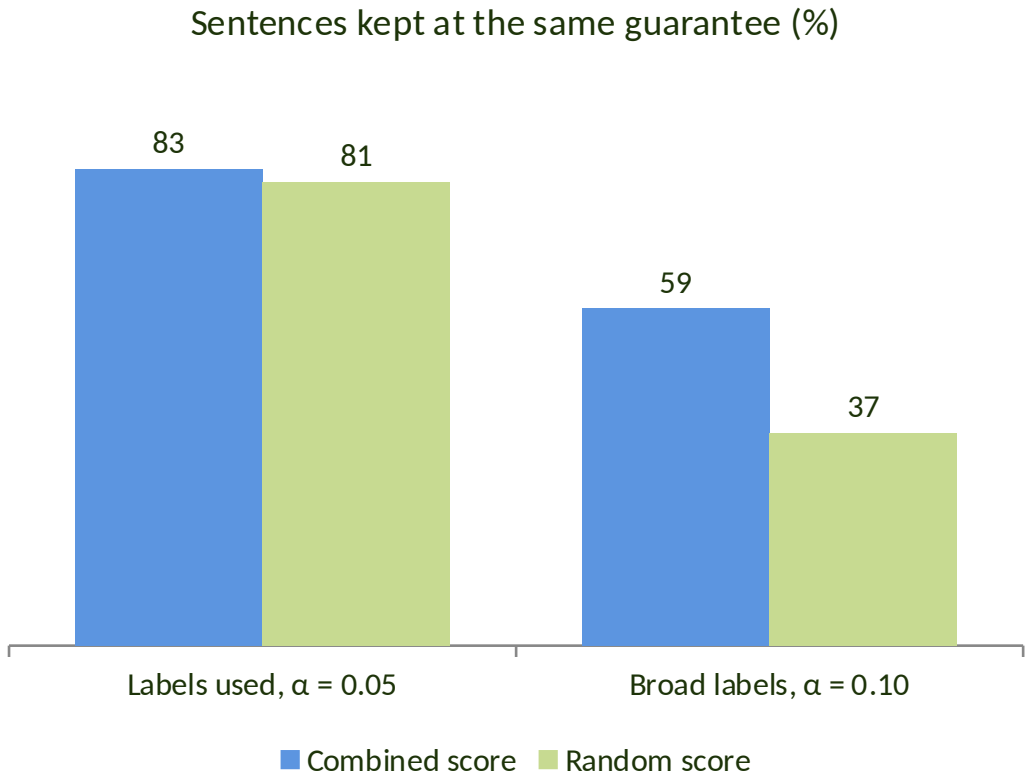
7 of 34

contradicted answers in a queue of 13% of drafts, with every held-out answer judged once (test split alone: 7 of 11)

Same rule, nothing re-tuned; the score never saw the calibration answers either. Answer-level AUROC 0.54 (95% CI 0.44–0.64): the guard tells the pharmacist how many drafts to read, not yet which.

The price depends on the detector





On the labels used, contradictions are rare (0.8% of sentences) and the score barely beats random: it learned mostly sentence position. On broad labels it keeps 59% against 37%, but most broad flags are false alarms.

One answer, sentence by sentence

Question: If I took trazodone when would be the best time to take it?

Decision: sent for review

~~Trazodone is usually best taken in the evening at bedtime because it commonly causes drowsiness...~~	0.969	× contradicts
For depression, doctors often prescribe a larger dose divided with the main portion at bedtime...	0.992	
~~It should be taken at the same time each day... often shortly after a meal or light snack...~~	0.986	× contradicts
~~Avoid alcohol and other sedatives near the dose because they worsen sleepiness, falls...~~	0.989	
~~Do not drive or operate machinery after taking it until you know how it affects you.~~	0.983	
If a dose is missed, the usual approach is to skip it if it is close to the next dose...	0.999	

Threshold $\tau = 0.9903$ at $\alpha = 0.05$. Both sentences that contradict the reference answer are dropped, but so are both safety warnings: the score can't tell a warning from an error. The two kept sentences go to a person (in LlamaFirewall, HUMAN_IN_THE_LOOP_REQUIRED).

Built on Meta's stack

Muse Spark 1.3

Every answer and resample through the Meta Model API: 4,914 generations. No logprobs, so the method is fully black-box.

LlamaFirewall

Llama Guard 3 1B screens the question first (self-harm → a person). Then the guard scans the answer: ship, review or block.

FAISS

Nearest-neighbour search over 15,575 MedlinePlus passages, the evidence behind each sentence's support score.

```
fw = LlamaFirewall(scanners={Role.USER: [register_question_scanner()],
                             Role.ASSISTANT: [register_scanner()]})
result = fw.scan_replay([UserMessage(content=question), AssistantMessage(content=answer)])
result.decision      # ALLOW / HUMAN_IN_THE_LOOP_REQUIRED / BLOCK
last_result()["certificate"]   # the guarantee that applies, in plain language
```

Limits, stated plainly, and what comes next

Limits

- A replay: answers were cached on Oct 26, before an automated account restriction (appeal pending)
- Labels are automatic (6 of 13 audited flags real), and a doctor wrote only 14% of the references
- The score learned mostly sentence position, so it can't yet pick which drafts get read (AUROC 0.54)
- MedlinePlus covers conditions, not drug labels
- The guarantee is marginal, assumes questions like MedLFQA's, and doesn't cover the reviewed subset

Next

- A score without position: evidence-first, trained on more human labels
- An answer-level score, so the queue targets errors (Conformal Alignment, Gui et al., 2024)
- Drug-label evidence (e.g. DailyMed) for medication questions
- Conditional guarantees per question type (Cherian et al., 2024)
- Fresh held-out answers once the appeal clears, with the analysis pre-registered
- A short paper with the full analysis

The guarantee held. The price is measured.

98.4%

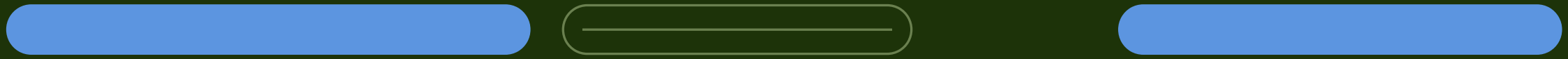
of held-out answers clean after filtering, against a 95% target (95.5% before)

4.7% ≤ 5%

shipped with a contradiction over 1,000 re-splits of all 573 held-out answers (5.9% without the guard)

1 in 6

drafts the pharmacist reads to keep that bound (17% on average)



Code, demo and full results are in the project repository.